Almost-Everywhere Affine Identifiability of iVAE

A measure-theoretic refinement of Theorem 1

Notation and Model Setup

Let $1 \le n \le d$, $m, k \ge 1$, $q = nk$, and $s \in \{\theta, \theta'\}$. Let $\mathcal{Z} \subseteq \mathbb{R}^n$ and $\mathcal{X} \subseteq \mathbb{R}^d$ be Borel, $\mathcal{U} \subseteq \mathbb{R}^m$, and

$$T_s(z) = \big(T_{s,i,j}(z_i)\big)_{i,j} \in \mathbb{R}^q, \qquad \lambda_s(u) = \big(\lambda_{s,i,j}(u)\big)_{i,j} \in \mathbb{R}^q.$$

Put $B_s(z) = \prod_i Q_{s,i}(z_i)$, $Z_s(u) = \prod_i Z_{s,i}(u)$, and

$$\begin{aligned} p_{s,u}(z) &= \frac{B_s(z)}{Z_s(u)} e^{\langle T_s(z), \lambda_s(u)\rangle} \\ &= \prod_{i=1}^{n} \left[\frac{Q_{s,i}(z_i)}{Z_{s,i}(u)} e^{\sum_{j=1}^{k} T_{s,i,j}(z_i)\lambda_{s,i,j}(u)} \right]. \end{aligned} \tag{1}$$

For injective $f_s : \mathbb{R}^n \to \mathbb{R}^d$ with $f_s(\mathcal{Z}) = \mathcal{X}$, write $g_s = (f_s|_{\mathcal{Z}})^{-1}$. Define

$$\begin{gathered} P_{s,u}(dz) = p_{s,u}(z)\, d\mathcal{L}^n|_{\mathcal{Z}}(z), \qquad t_s = T_s \circ g_s, \\ \nu_{s,u} = (f_s|_{\mathcal{Z}})_{\#} P_{s,u}, \qquad \bar{\nu}_{s,u} = \iota_{\#}\nu_{s,u}, \qquad \iota : \mathcal{X} \hookrightarrow \mathbb{R}^d. \end{gathered}$$

For a common independent additive noise law η, let

$$O_{s,\eta}(u) = \bar{\nu}_{s,u} * \eta. \tag{2}$$

For a fixed measure μ on $\mathcal{X}$, set

$$\theta \sim_A^{\mu} \theta' \iff \exists A \in \mathrm{GL}_q(\mathbb{R}),\ c \in \mathbb{R}^q : \quad t_\theta = A t_{\theta'} + c \quad \mu\text{-a.e.}$$

This is an equivalence relation, by the identity, inverse, and composition of affine maps on common full-μ-measure sets.

Theorem 1 (Corrected a.e. identifiability)**.** *Assume the following.*

(1) ***Injective mixing and measurability.*** (paper assumption (ii); explicit measurable-image interface) *For both models, $f_s : \mathbb{R}^n \to \mathbb{R}^d$ is injective, $f_s(\mathcal{Z}) = \mathcal{X}$, and $f_s|_{\mathcal{Z}} : \mathcal{Z} \to \mathcal{X}$ and $g_s : \mathcal{X} \to \mathcal{Z}$ are measurable.*

(2) ***Valid product-form priors.*** (paper equation (7); explicit probability-model validity) *For both models,*

$$Q_{s,i}(v) \ge 0 \quad (\forall i, v), \qquad 0 < Z_{s,i}(u) < \infty \quad (\forall i, u),$$

and, for every $u \in \mathcal{U}$, $p_{s,u}$ is measurable with $\int_{\mathcal{Z}} p_{s,u}(z)\, dz = 1$.

(3) ***Full-frequency noise condition.*** (strengthened assumption (i)) *The measure η is a probability measure on $\mathbb{R}^d$ and*

$$\mathcal{L}^d\{\omega \in \mathbb{R}^d : \varphi_\eta(\omega) = 0\} = 0, \qquad \varphi_\eta(\omega) = \int e^{\mathrm{i}\langle \omega, y\rangle}\, d\eta(y). \tag{3}$$

(4) ***Auxiliary variation.*** (paper assumption (iv)) *There exist pairwise distinct $u_0, \dots, u_q \in \mathcal{U}$ such that*

$$L = [\lambda_\theta(u_1) - \lambda_\theta(u_0) \ \cdots \ \lambda_\theta(u_q) - \lambda_\theta(u_0)], \qquad \det L \neq 0. \tag{4}$$

(5) ***Essential affine nondegeneracy of the true pivot prior.*** (additional assumption for Step III; related to assumption (iii)) *For every* $v \in \mathbb{R}^q$, $b \in \mathbb{R}$,

$$[\,\langle v, T_\theta(z)\rangle = b \quad P_{\theta,u_0}\text{-a.e.}\,] \implies v = 0. \tag{5}$$

This condition is assumed, not deduced from the paper's assumption (iii).

If

$$O_{\theta,\eta}(u_r) = O_{\theta',\eta}(u_r) \qquad (r = 0, \dots, q), \tag{6}$$

then $\sigma := \nu_{\theta,u_0} = \nu_{\theta',u_0}$ *and*

$$\exists A \in \mathrm{GL}_q(\mathbb{R}),\ c \in \mathbb{R}^q : \quad t_\theta(x) = A t_{\theta'}(x) + c \quad \sigma\text{-a.e.} \tag{7}$$

Consequently,

$$\mu \ll \sigma \implies \theta \sim_A^\mu \theta'. \tag{8}$$

The same A, c *apply to every such* μ*, including all* $\nu_{s,u}$.

Proof.

Step I. For $r = 0, \dots, q$, the convolution identity and (6) give

$$\varphi_{\bar\nu_{\theta,u_r}} \varphi_\eta = \varphi_{\bar\nu_{\theta',u_r}} \varphi_\eta.$$

By (3), continuity of characteristic functions, and Fourier uniqueness for finite measures,

$$\begin{aligned} &\varphi_{\bar\nu_{\theta,u_r}} = \varphi_{\bar\nu_{\theta',u_r}} \quad \mathcal{L}^d\text{-a.e.} \\ &\implies \varphi_{\bar\nu_{\theta,u_r}} = \varphi_{\bar\nu_{\theta',u_r}} \quad \text{everywhere} \\ &\implies \bar\nu_{\theta,u_r} = \bar\nu_{\theta',u_r}. \end{aligned}$$

Here continuity upgrades a.e. equality since every nonempty open subset of $\mathbb{R}^d$ has positive Lebesgue measure. The inclusion ι is a measurable embedding; hence

$$\nu_{\theta,u_r} = \nu_{\theta',u_r} \qquad (r = 0, \dots, q). \tag{9}$$

In particular, $\sigma = \nu_{\theta,u_0} = \nu_{\theta',u_0}$ is a probability measure.

Step II. Fix $s \in \{\theta, \theta'\}$ and $u \in \mathcal{U}$. By (1),

$$p_{s,u}(z) = 0 \iff B_s(z) = 0 \iff p_{s,u_0}(z) = 0. \tag{10}$$

Define the nonnegative measurable ratio

$$r_{s,u}(z) = \begin{cases} p_{s,u}(z)/p_{s,u_0}(z), & p_{s,u_0}(z) > 0, \\ 0, & p_{s,u_0}(z) = 0. \end{cases} \tag{11}$$

Using (10),

$$p_{s,u_0} r_{s,u} = p_{s,u} \quad \text{on } \mathcal{Z}, \tag{12}$$

and therefore

$$P_{s,u} = r_{s,u}\, P_{s,u_0}. \tag{13}$$

Let $R_{s,u} = r_{s,u} \circ g_s$. For every measurable $E \subseteq \mathcal{X}$,

$$\nu_{s,u}(E) = \int_{(f_s|_\mathcal{Z})^{-1}(E)} r_{s,u}\, dP_{s,u_0} = \int_E R_{s,u}\, d\nu_{s,u_0}.$$

Thus

$$\nu_{s,u} = R_{s,u}\,\nu_{s,u_0}, \qquad \nu_{s,u} \ll \nu_{s,u_0}. \tag{14}$$

Since $P_{s,u_0}\{p_{s,u_0} = 0\} = 0$, cancellation in (11) gives

$$r_{s,u}(z) = \frac{Z_s(u_0)}{Z_s(u)} e^{\langle T_s(z), \lambda_s(u) - \lambda_s(u_0)\rangle} > 0 \quad P_{s,u_0}\text{-a.e.}$$

Set $a_s(u) = \log Z_s(u)$ and $\Delta\lambda_s(u) = \lambda_s(u) - \lambda_s(u_0)$. Taking logarithms and pushing forward,

$$\log R_{s,u}(x) = \langle t_s(x), \Delta\lambda_s(u)\rangle - [a_s(u) - a_s(u_0)] \quad \nu_{s,u_0}\text{-a.e.} \tag{15}$$

From (9) and (14),

$$R_{\theta,u_r}\,\sigma = R_{\theta',u_r}\,\sigma.$$

Uniqueness of densities with respect to the same finite measure yields

$$R_{\theta,u_r} = R_{\theta',u_r} \quad \sigma\text{-a.e.} \tag{16}$$

Combining (15)–(16),

$$\begin{aligned}\langle t_\theta(x), \Delta\lambda_\theta(u_r)\rangle - [a_\theta(u_r) - a_\theta(u_0)] = \langle t_{\theta'}(x), \Delta\lambda_{\theta'}(u_r)\rangle \\ - [a_{\theta'}(u_r) - a_{\theta'}(u_0)] \quad \sigma\text{-a.e.}\end{aligned} \tag{17}$$

Intersecting finitely many full-σ-measure sets makes these identities simultaneous for $r = 1, \ldots, q$. With

$$\begin{aligned} L' &= [\Delta\lambda_{\theta'}(u_1) \;\cdots\; \Delta\lambda_{\theta'}(u_q)], \\ b_r &= [a_\theta(u_r) - a_\theta(u_0)] - [a_{\theta'}(u_r) - a_{\theta'}(u_0)], \end{aligned}$$

equation (17) becomes

$$L^\top t_\theta = (L')^\top t_{\theta'} + b \quad \sigma\text{-a.e.} \tag{18}$$

Since $\det L \neq 0$, set

$$A = (L^\top)^{-1}(L')^\top, \qquad c = (L^\top)^{-1} b. \tag{19}$$

Then

$$t_\theta = A t_{\theta'} + c \quad \sigma\text{-a.e.} \tag{20}$$

Step III. Using $\sigma = (f_\theta|_{\mathcal{Z}})_\# P_{\theta,u_0}$ and $t_\theta \circ f_\theta|_{\mathcal{Z}} = T_\theta|_{\mathcal{Z}}$, assumption (5) gives

$$\begin{aligned}[\,\langle v, t_\theta\rangle = b \quad \sigma\text{-a.e.}\,] &\Longrightarrow [\,\langle v, T_\theta\rangle = b \quad P_{\theta,u_0}\text{-a.e.}\,] \\ &\Longrightarrow v = 0.\end{aligned}$$

If A were singular, choose $0 \neq v \in \ker A^\top$. By (20),

$$\langle v, t_\theta\rangle = \langle A^\top v, t_{\theta'}\rangle + \langle v, c\rangle = \langle v, c\rangle \quad \sigma\text{-a.e.},$$

forcing $v = 0$, a contradiction. Thus $A \in \mathrm{GL}_q(\mathbb{R})$.

For $\mu \ll \sigma$, the exceptional set in (20) is also μ-null, proving (8) with the same A, c. Finally, (14) implies $\nu_{s,u} \ll \nu_{s,u_0} = \sigma$ for every s, u. □

Scope. Only injectivity on $\mathcal{Z}$ is used; no ambient surjectivity or $n = d$ assumption is required. The conclusion is a.e., not pointwise. The full-frequency condition and (5) are explicit corrections to the original theorem of [1].

References

[1] I. Khemakhem, D. P. Kingma, R. Monti, and A. Hyvärinen. *Variational Autoencoders and Nonlinear ICA: A Unifying Framework*. arXiv:1907.04809v4, 2020. Theorem 1 and Appendix B.2. `https://arxiv.org/abs/1907.04809v4`.